BTP-I Mid-Semester Project Summary

Dynamic Truck–Drone Collaborative Delivery with Flexible Launch Positions

by

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***Abstract*—*Truck–drone collaborative delivery with flexible launch positions has been studied as a static planning problem and, more recently, as a dynamic problem solved through a learned dispatch policy; separately, dynamic re-planning of an already-executing delivery route has been studied for single-vehicle attended-delivery systems using local-search heuristics. This project first reproduces the static formulation of Han, Chi, and Sun [1] as a validated baseline, then investigates a computational mechanism for requests that arrive while a plan is already executing that differs from both dynamic references: localized modification of the unexecuted portion of the plan under customer-specific delay tolerances, in contrast to the Markov-decision-process/reinforcement-learning policy of Cui et al. [2] and the whole-route re-optimization used by Chakraborty et al. [3]. This summary states the problem, positions it relative to the three references, and reports current implementation status.*

I. Motivation and Research Context

Combining a road-based truck with air-based drones lets the two vehicles serve different customers in parallel. Han et al. [1] formalize a static version in which a drone may launch from a flexible point along the truck's route, not only from a customer stop, and is recovered at a truck-service node or the ending depot. The full customer set is known when the plan is built, and the objective is total customer waiting time, solved by an MILP-guided VNS heuristic.

Cui et al. [2] address a genuinely dynamic truck–drone problem: requests for a fleet of multiple trucks and drones arrive randomly, each with its own time window and a delay penalty, and both launch and retrieval positions are chosen en-route. They cast the real-time decisions as a Markov decision process and solve them with a reinforcement-learning deep Q-network policy that decides, per request, whether and by which vehicle it is served.

A separate line of work addresses dynamic re-planning without drones. Chakraborty, Darbhe, and Sarmah [3] study attended home delivery for India's public distribution system with a single vehicle per depot. A newly arriving request is checked against an already-planned day's route and, if accepted, an iterated-local-search/adaptive-large- neighborhood-search metaheuristic is rerun over the unexecuted route.

This project poses the dynamic question differently. Rather than learning a policy over arrival states or re-optimizing the whole remaining route, Stage 2 asks how an already-executing truck–drone plan produced by Stage 1 should be locally repaired: the completed portion is held fixed, changes are confined to a bounded region around the new request, and previously scheduled customers are required to remain within their individual delay tolerances.

II. Static Baseline and Dynamic Problem

Stage 1 reproduces Han et al.'s static problem: one truck, multiple drones, a known customer set, flexible launch positions (FLPs) generated along the arcs actually traversed by the truck, drone recovery restricted to a truck-service node or the ending depot, drone-endurance and truck/drone synchronization constraints, and a total-waiting-time objective. Flexible recovery is therefore not introduced, unlike the en-route retrieval considered by Cui et al.

Stage 2 begins once a plan S produced by Stage 1 has been validated and is executing. When a new request c_{new} arrives, the executed portion of S is frozen. A candidate modification S' may only act on the remainder and is confined to an affected region around c_{new}, such as nearby customers, the local route segment, idle drones, and nearby FLPs. Flexible launch positions are retained; flexible recovery is not introduced.

For each existing customer i, with planned delivery time T_i and modified delivery time T_i', define the delivery delay as

$$d_i = T_i' - T_i, \qquad d_i \le T_i^h, \tag{1}$$

where T_i^h is the customer's allowable delay tolerance relative to its own scheduled time.

Candidates are compared using the change in total waiting time ΔW, whether c_{new} is served, the number of existing customers affected, tolerance violations, and computation time. A working scalarization is

$$\min \quad \Delta W - \alpha N_{\text{new}} + M \sum_{i \in C_{\text{old}}} v_i, \tag{2}$$

where

$$v_i = \max\left(0, d_i - T_i^h\right). \tag{3}$$

Here α and M are design parameters to be calibrated from baseline results; they are not established values.

III. Proposed Mathematical Formulation

Let S denote the currently executing plan and S' a candidate modified plan. The Stage-2 decision is to select a feasible local modification of S that accommodates the new request while limiting the disruption to existing customers.

The proposed formulation is represented compactly as

$$\begin{aligned} \min_{S'} \quad & \Delta W - \alpha N_{\text{new}} + M \sum_{i \in C_{\text{old}}} v_i \\ \text{s.t.} \quad & S'_{\text{executed}} = S_{\text{executed}}, \\ & S'_{\text{changed}} \subseteq R(c_{\text{new}}), \\ & d_i \le T_i^h, \qquad i \in C_{\text{old}}, \\ & \text{truck–drone synchronization}, \\ & \text{drone endurance}, \\ & \text{drone fleet capacity}, \\ & \text{route and service feasibility}. \end{aligned} \tag{4}$$

Here $R(c_{\text{new}})$ denotes the affected region selected around the new request. Its exact definition and size are research variables to be investigated experimentally. The formulation is therefore a working Stage-2 model rather than a finalized mathematical program.

The principal design distinction is that the completed portion of the original plan is immutable, while only a bounded portion of the unexecuted plan may be changed. Candidate plans are evaluated using the same deterministic timing and feasibility engine developed for Stage 1.

IV. Methodology and Implementation

Stage-1 pipeline (implemented): instance → truck-only TSP → k-means clustering → FLP generation on traversed arcs → greedy drone-trip construction with temporal fleet-capacity checking → initial solution. A VNS driver with eight local-search and four shake neighborhoods, following Han et al., is designed to refine the solution but is not yet fully implemented.

Every stage calls one timing/feasibility engine that is the sole place computing arrival times, synchronization, endurance, fleet capacity, feasibility, and the objective. Thus, Stage 2 will later evaluate candidate modifications using the same deterministic rules.

Stage-2 pipeline (planned): executing plan → new request → freeze committed portion → identify affected region → generate local modifications (truck insertion, drone assignment, launch-position change, local reordering) → evaluate candidates through the Stage-1 timing engine → select a feasible modification within tolerance, or reject.

Neighborhood-based and clustering-based affected-region definitions are candidate strategies to be compared experimentally. Unlike the day-level acceptance test of Chakraborty et al. [3], the decision unit here is a bounded region rather than the entire remaining route.

V. Current Progress

TABLE I. Current Implementation Status

Component	Status
Core data model, geometry & travel-time layer	**Done**
Flexible launch point generation	**Done**
Timing & feasibility engine	**Done**
Initial-solution construction	**Done**
VNS driver and neighborhoods	*In progress*
Experiment runner & validation vs. Han et al.	Planned
Stage-2 dynamic layer	Planned

The completed components are exercised by a regression suite of 339 passing assertions covering the data model, geometry, FLP generation, timing/feasibility, and initial-solution construction. The VNS driver and downstream components are not yet covered because they are still under development.

VI. Positioning Relative to Reference Studies

TABLE II. Feature Comparison with Reference Studies

	SP	DR	TD	FL	FR	RL	LR	DT
Han et al. [1]	✓	–	✓	✓	–	–	–	–
Cui et al. [2]	✓	✓	✓	✓	✓	✓	–	–
Chakraborty et al. [3]	✓	✓	–	–	–	–	WR	–
This work	✓ (S1)	✓ (S2)	✓	✓	–	–	✓	✓

SP: static planning; **DR**: dynamic requests; **TD**: truck–drone coordination; **FL**: flexible launch; **FR**: flexible recovery; **RL**: learned RL/MDP policy; **LR**: localized repair; **WR**: whole remaining route; **DT**: per-customer delay tolerance. For **This work**, **(S1)** and **(S2)** indicate the project stage: S1 is the static Stage-1 baseline, already implemented and tested, while S2 is the proposed dynamic Stage-2 extension.

VII. Research Direction and Remaining Work

The immediate task is finishing the VNS driver and its twelve neighborhoods, followed by an experiment runner that reproduces Han et al.'s synthetic instances and checks objective values and runtime against their reported results. Mismatches arising from unspecified paper details will be distinguished from implementation errors.

Only after this baseline is validated will Stage 2 be implemented. The next components are the event layer, affected-region strategies, and local-modification generator, followed by single-request experiments. These experiments will compare region definitions and sizes using ΔW, tolerance violations, number of affected customers, and computation time.

The Stage-2 direction differs from Cui et al.'s trained dispatch policy because no policy is learned across episodes. It also differs from Chakraborty et al. [3] because acceptance and modification are confined to a bounded region around the arriving request rather than the whole remaining route.

VIII. Conclusion

Stage 1 reproduces Han et al.'s static flexible-launch problem, with the core computational pipeline implemented and independently tested; VNS refinement and paper-level validation remain. Stage 2 will investigate a different dynamic mechanism: localized modification of an already-executing truck–drone plan under customer-specific delay tolerances. The objective is to accommodate newly arriving requests while limiting disruption to the existing plan, using the validated Stage-1 timing and feasibility framework as its computational foundation.

References

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[3] S. Chakraborty, K. Darbhe, and S. Sarmah, "Attended home delivery in Indian public distribution system: an iterated local search approach," *Journal of Modelling in Management*, vol. 16, no. 4, pp. 1279–1301, 2021, doi: 10.1108/JM2-06-2020-0148.

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